```python
#Analysis start
#Top revenue contributing model
df.groupby('model')['revenue_eur'].sum().sort_values(ascending=False)
```

```
model
X7          39577415992
iX          33420142373
X5          30845881310
5 Series    26686164121
i4          25381949369
X3          22480326576
3 Series    20236075907
MINI        16977726266
Name: revenue_eur, dtype: int64
```

```python
# 2: Highest revenue efficiency model
df.groupby('model')['revenue_per_unit'].mean().sort_values(ascending=False)
```

```
model
X7          92158.020833
iX          75053.500000
X5          68913.312500
i4          64853.250000
5 Series    61879.020833
X3          58023.895833
3 Series    47836.541667
MINI        41894.125000
Name: revenue_per_unit, dtype: float64
```

```python
# EV models with strongest sales alignment
df.groupby('bev_category')['units_sold'].mean()
```

```
C:\Users\ahama\AppData\Local\Temp\ipykernel_15612\4282757385.py:2: FutureWarning: The default of
e in a future version of pandas. Pass observed=False to retain current behavior or observed=True
  df.groupby('bev_category')['units_sold'].mean()
```

```
bev_category
Low EV Demand                  NaN
Moderate EV Demand     8847.898551
High EV Demand         8525.527778
Very High EV Demand            NaN
Name: units_sold, dtype: float64
```

```python
#Fuel sensitivity impact
df.groupby('model')['fuel_sensitivity_score'].mean().sort_values(ascending=False)
```

```
model
X5          12634.571250
iX          12571.330417
5 Series    12192.510000
X7          12138.878542
3 Series    11945.968333
MINI        11442.978958
i4          11025.084167
X3          10934.202708
Name: fuel_sensitivity_score, dtype: float64
```

```sql
-- Q1 Total revenue generated by each model
SELECT model, SUM(revenue_eur) AS total_revenue
FROM bmw_sales
GROUP BY model
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

model	total_revenue
X7	39577415992
iX	33420142373
X5	30845881310
5 Series	26686164121
i4	25381949369
X3	22480326576
3 Series	20236075907
MINI	16977726266

```sql
-- Q2 Total units sold by region
SELECT region, SUM(units_sold) AS total_units
FROM bmw_sales
GROUP BY region
ORDER BY total_units DESC;
```

Result Grid | Filter Rows: | Export: | Wrap

region	total_units
China	887189
USA	873519
Europe	824827
RestOfWorld	777242

```sql
-- Q3 Average selling price by model
SELECT model, AVG(avg_price_eur) AS avg_price
FROM bmw_sales
GROUP BY model
ORDER BY avg_price DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

model	avg_price
X7	92158.0208
iX	75053.5000
X5	68913.3125
i4	64853.2500
5 Series	61879.0208
X3	58023.8958
3 Series	47836.5417
MINI	41894.1250

```sql
-- Q4 Highest premium share region
SELECT region, AVG(premium_share) AS premium_share_avg
FROM bmw_sales
GROUP BY region
ORDER BY premium_share_avg DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

region	premium_share_avg
China	17.73333333333335
USA	17.398333333333362
Europe	17.214999999999982
RestOfWorld	5.5575

```sql
-- Q5 EV share by region
SELECT region, AVG(bev_share) AS avg_bev_share
FROM bmw_sales
GROUP BY region
ORDER BY avg_bev_share DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

region	avg_bev_share
Europe	0.19504166666666659
China	0.1941770833333334
USA	0.19415625000000006
RestOfWorld	0.1937291666666667

```sql
-- Q6 GDP growth effect on revenue
SELECT region, AVG(gdp_growth) AS avg_gdp, SUM(revenue_eur) AS revenue
FROM bmw_sales
GROUP BY region
ORDER BY revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

region	avg_bev_share
Europe	0.19504166666666659
China	0.1941770833333334
USA	0.19415625000000006
RestOfWorld	0.1937291666666667

```sql
-- Q7 Fuel price impact on sales
SELECT fuel_price_index, SUM(units_sold) AS total_sales
FROM bmw_sales
GROUP BY fuel_price_index
ORDER BY fuel_price_index;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

fuel_price_index	total_sales
1.29	71088
1.31	250976
1.32	72019
1.33	286626
1.34	514040
1.35	520390
1.36	548476
1.37	356550
1.38	348237
1.39	197132
1.4	68870
1.41	128373

```sql
-- Q8 Top 5 models by revenue
SELECT model, revenue_eur
FROM bmw_sales
ORDER BY revenue_eur DESC
LIMIT 5;
```

Result Grid | Filter Rows:

model	revenue_eur
X7	1433481712
X7	1364422508
X7	1362720307
X7	1333600992
X7	1255313682

```sql
-- Q9 Monthly sales trend
SELECT month, SUM(units_sold) AS total_sales
FROM bmw_sales
GROUP BY month
ORDER BY month;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

month	total_sales
1	267230
2	251759
3	351705
4	260838
5	255299
6	282550
7	303552
8	260474
9	306776
10	237864
11	285728
12	299002

```sql
-- Q10 Region wise premium revenue
SELECT region, SUM(revenue_eur) AS premium_revenue
FROM bmw_sales
GROUP BY region
ORDER BY premium_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

region	premium_revenue
China	57155301885
USA	55388019620
Europe	53324850050
RestOfWorld	49737510359

```sql
-- Q11 Highest priced models
SELECT model, avg_price_eur
FROM bmw_sales
ORDER BY avg_price_eur DESC;
```

Result Grid | Filter Rows: | Export:

model	avg_price_eur
X7	93853
X7	93771
X7	93598
X7	93587
X7	93538
X7	93480
X7	93460
X7	93319
X7	93295
X7	93284
X7	93204
X7	93098

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```sql
-- Q12 Revenue and units together
SELECT model, SUM(units_sold), SUM(revenue_eur)
FROM bmw_sales
GROUP BY model;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

model	SUM(units_sold)	SUM(revenue_eur)
3 Series	423608	20236075907
5 Series	431863	26686164121
X3	387534	22480326576
X5	447715	30845881310
X7	429678	39577415992
i4	391237	25381949369
iX	445304	33420142373
MINI	405838	16977726266